Overlap inflation in EEG forecasting

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Abstract

The abstract paragraph should be indented ½ inch (3 picas) on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Introduction

Electroencephalography (EEG) records the small electrical voltages produced by populations of neurons through electrodes placed on the scalp. Because the measurement is noninvasive, resolves activity on the order of milliseconds, and is inexpensive relative to other means of observing the brain, it is the signal most widely used for brain–computer interfaces, real-time neurotechnology, and continuous monitoring of brain state. Each of those applications is a closed loop, and a closed loop is limited by delay: a system that can only react to an event it has already observed acts too late. Forecasting the signal a short time ahead is what removes that delay, and it would enable stimulation delivered before a seizure spreads rather than after it is detected, prosthetic control without the lag users cite when they abandon a device, and interfaces that respond as a user acts rather than once they have. The same promise motivates current interest in foundation models for EEG, which aim to replace a bespoke model per clinic or per task with one pretrained system that transfers. Whether such models can forecast at all bears directly on what that programme can support.

What makes this hard is the measurement. Because the electrodes sit outside the skull, each one records the summed activity of many millions of neurons after it has been blurred and attenuated by intervening tissue, which yields poor spatial resolution and a low ratio of brain activity to everything else. Muscle tension, eye movement, and line interference routinely exceed the activity of interest. The signal is also non-stationary: its statistics drift within a recording and differ substantially between people, so a representation fitted to one session need not hold on the next. The dominant response has been to pretrain a large model on many hours of unlabelled recordings and transfer the learned representation to a downstream task, on the argument that scale supplies the invariances that hand-designed features lack. That approach has produced strong reported numbers, but recent independent evaluations find frozen pretrained representations often fail to beat well-tuned classical baselines, which suggests some reported gains may owe more to how they are measured than to what was learned.

Forecasting is especially exposed to this, because the standard way of scoring it can manufacture the appearance of skill. When a model is scored on a window that overlaps the window it was given as input, it can score well by copying its input forward, and published protocols routinely permit that overlap; the effect is large enough to make a model appear nearly perfect on data containing no predictable structure whatsoever. Deep models are also frequently compared against baselines weaker than a properly tuned linear map, and trained and tested on recordings from the same person. Nothing in such a protocol requires a model to generalise to a new subject, or to a sample that genuinely lies in the future.

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This paper does three things. We derive exactly how much overlap inflates the two scores reported in this literature, mean squared error and correlation, and show the correction is not the same for both: for squared error the inflation is a closed-form ratio of window length to horizon, W/H, which we verify is never violated across 286 configurations, while correlation carries an additional shrinkage term that makes W/H understate the inflation on 223 of those cells, always in the flattering direction. The sharpest case is white noise scored at $W{=}32$, $H{=}1$, which reports $r = 0.984$ where the true forecast correlation is 0.014. Applying this to a $100{,}000$-step recurrent model on BNCI2014_001, its reported $r = 0.972$ is retracted for a stronger reason than its MSE is. We then specify an evaluation in which splits are frozen before windowing, subjects are held out, and the scored sample is never in the input, with persistence, ridge, and mean as mandatory baselines. Finally we run our own model through it. After subtracting what ridge can predict and stripping narrowband oscillations, a residual recurrent stack retains a 10.5–17.6% MSE advantage over a per-fold-tuned 127-lag ridge on 78-subject Sleep-EDF leave-one-subject-out, with per-horizon bootstrap confidence intervals excluding zero and the advantage holding on 72–78 of 78 held-out subjects. We treat this as a powered preliminary positive rather than a result: one seed, one channel of one corpus, an unreproduced normalisation change, and a 9-subject BNCI arm on which the same protocol cannot separate the model from tuned ridge at any horizon. What we claim is the protocol and the retraction.

2 Submission of papers to NeurIPS 2026

Please read the instructions below carefully and follow them faithfully.

2.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

2.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

`https://neurips.cc`.

The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for LaTeX 2_ε. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported.**

The LaTeX style file contains three optional arguments:

- `final`, which creates a camera-ready copy,
- `preprint`, which creates a preprint for submission to, e.g., arXiv,
- `nonatbib`, which will not load the `natbib` package for you in case of package clash.

Preprint option If you wish to post a preprint of your work online, e.g., on arXiv, using the NeurIPS style, please use the `preprint` option. This will create a nonanonymized version of your work with the text "Preprint. Work in progress." in the footer. This version may be distributed as you see fit, as long as you do not say which conference it was submitted to. Please **do not** use the `final` option, which should **only** be used for papers accepted to NeurIPS.

At submission time, please omit the `final` and `preprint` options. This will anonymize your submission and add line numbers to aid review. Please do *not* refer to these line numbers in your paper as they will be removed during generation of camera-ready copies.

The file `neurips_2026.tex` may be used as a "shell" for writing your paper. All you have to do is replace the author, title, abstract, and text of the paper with your own.

The formatting instructions contained in these style files are summarized in Sections 3, 4, and 5 below.

3 General formatting instructions

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing (leading) of 11 points. Times New Roman is the preferred typeface throughout, and will be selected for you by default. Paragraphs are separated by ½ line space (5.5 points), with no indentation.

The paper title should be 17 point, initial caps/lower case, bold, centered between two horizontal rules. The top rule should be 4 points thick and the bottom rule should be 1 point thick. Allow ¼ inch space above and below the title to rules. All pages should start at 1 inch (6 picas) from the top of the page.

For the final version, authors' names are set in boldface, and each name is centered above the corresponding address. The lead author's name is to be listed first (left-most), and the co-authors' names (if different address) are set to follow. If there is only one co-author, list both author and co-author side by side.

Please pay special attention to the instructions in Section 5 regarding figures, tables, acknowledgments, and references.

4 Headings: first level

All headings should be lower case (except for first word and proper nouns), flush left, and bold.

First-level headings should be in 12-point type.

4.1 Headings: second level

Second-level headings should be in 10-point type.

4.1.1 Headings: third level

Third-level headings should be in 10-point type.

Paragraphs There is also a `\paragraph` command available, which sets the heading in bold, flush left, and inline with the text, with the heading followed by 1 em of space.

5 Citations, figures, tables, references

These instructions apply to everyone.

5.1 Citations within the text

The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as long as you maintain internal consistency. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

The documentation for `natbib` may be found at

```
http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf
```

Of note is the command `\citet`, which produces citations appropriate for use in inline text. For example,

```
\citet{hasselmo} investigated\dots
```

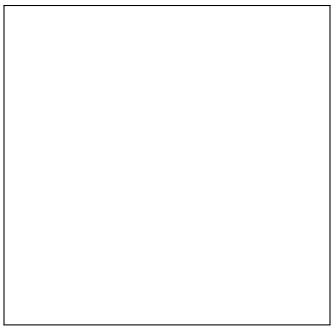


Figure 1: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

produces

Hasselmo, et al. (1995) investigated...

If you wish to load the `natbib` package with options, you may add the following before loading the `neurips_2026` package:

```
\PassOptionsToPackage{options}{natbib}
```

If `natbib` clashes with another package you load, you can add the optional argument `nonatbib` when loading the style file:

```
\usepackage[nonatbib]{neurips_2026}
```

As submission is double blind, refer to your own published work in the third person. That is, use "In the previous work of Jones et al. [4]," not "In our previous work [4]." If you cite your other papers that are not widely available (e.g., a journal paper under review), use anonymous author names in the citation, e.g., an author of the form "A. Anonymous" and include a copy of the anonymized paper in the supplementary material.

5.2 Footnotes

Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number[1] in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).

Note that footnotes are properly typeset *after* punctuation marks.[2]

5.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

5.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 1.

[1] Sample of the first footnote.

[2] As in this example.

Table 1: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

	Part	
Name	Description	Size (μm)
Dendrite	Input terminal	$\approx$100
Axon	Output terminal	$\approx$10
Soma	Cell body	up to 10^6

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

`https://www.ctan.org/pkg/booktabs`

This package was used to typeset Table 1.

5.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using $$ anyway; see `https://tex.stackexchange.com/questions/503/why-is-preferable-to` and `https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath` for more information.)

5.6 Final instructions

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes. Please note that pages should be numbered.

6 Preparing PDF files

Please prepare submission files with paper size "US Letter," and not, for example, "A4."

Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or Embedded TrueType fonts. Here are a few instructions to achieve this.

- You should directly generate PDF files using `pdflatex`.
- You can check which fonts a PDF files uses. In Acrobat Reader, select the menu Files>Document Properties>Fonts and select Show All Fonts. You can also use the program `pdffonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- `xfig` "patterned" shapes are implemented with bitmap fonts. Use "solid" shapes instead.
- The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS Fonts:

  ```
  \usepackage{amsfonts}
  ```

  followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also use the following workaround for reals, natural and complex:

  ```
  \newcommand{\RR}{I\!\!R} %real numbers
  \newcommand{\Nat}{I\!\!N} %natural numbers
  \newcommand{\CC}{I\!\!\!\!C} %complex numbers
  ```

Note that `amsfonts` is automatically loaded by the `amssymb` package.

If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

6.1 Margins in LaTeX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below:

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

See Section 4.4 in the graphics bundle documentation (`http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf`)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command when necessary.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

Note: Think of the appendix as "optional reading" for reviewers. The paper must be able to stand alone without the appendix; for example, adding critical experiments that support the main claims to an appendix is inappropriate.